

CYBER SECURITY

- 1. Explain the different types of Information Systems and discuss the major threats to Information Systems in modern organisations. How does Information Assurance help in mitigating these threats?**
- 2. Describe in detail the TCP/IP Protocol Suite. Explain how each layer contributes to the security and functioning of computer networks with suitable examples.**
- 3. What is Application Security? Discuss database security, email security, and internet security in depth. Also explain the importance of data backups, archival storage, and safe disposal of data.**
- 4. Explain different types of malicious software—viruses, worms, Trojan horses, bombs, trapdoors, spoofing, email viruses, and macro viruses. Give real-world examples and explain their prevention mechanisms.**
- 5. Describe in detail the various Security Technologies such as Firewalls, VPNs, Intrusion Detection Systems, and Access Control mechanisms. Compare their roles in organizational cybersecurity.**
- 6. What are the major security threats to E-Commerce systems? Explain the working and security of Electronic Payment Systems, Digital Signature, Public Key Cryptography, Credit and Debit Card transactions.**
- 7. Discuss the concept of Information Security Governance and Risk Management. Explain how secure information systems are developed, including security issues in hardware, data storage, and physical security of IT assets.**
- 8. What are Security Policies? Explain WWW policies, Email security policies, corporate security policies, their review process, and publishing/notification requirements in detail.**
- 9. Discuss the major provisions of the IT Act 2000 related to Cyber Crimes and Cyber Laws in India. Also explain Copyright Law, Software Licensing, Semiconductor Law, Patent Law and Intellectual Property Rights in detail.**
- 10. Explain in detail the steps involved in password protection in Microsoft Word on different operating systems. Also discuss password cracking techniques, their types, tools used, and how to prevent password-based hacking attacks.**